

Latency Report

Goal: measure the raw STT -> LLM -> TTS voice pipeline, identify where latency comes from, then compare a deterministic speech-directive optimization.

Run overview

Run	Streaming	Turns	Tool turns	End-to-heard median / p95	Tool end-to-heard median / p95	Tool LLM total median / p95	Tool TTS complete median / p95
No streaming - LLM mode	Off	9	6	6.61s / 13.95s	7.20s / 13.95s	2.85s / 8.51s	2.14s / 2.93s
No streaming - Directive mode	Off	9	4	6.19s / 8.76s	5.95s / 6.78s	1.84s / 1.89s	1.99s / 3.07s
Streaming - Directive mode	Pending	-	-	-	-	-	-

Key findings

- End-of-turn detection was stable in both non-streaming runs at about 0.68s median, so it did not explain the mode difference.
- Local tool execution was effectively zero because tools run as local backend functions over sample data.
- In the clean comparison, TTS was not the main differentiator: tool-turn complete synthesis was 2.14s in LLM mode vs 1.99s in directive mode.
- The model layer was the main optimized step: directive mode removed the post-tool LLM response pass, reducing tool-turn LLM total from 2.85s median / 8.51s p95 to 1.84s median / 1.89s p95.
- Determinism improved: in LLM mode the model can paraphrase or choose a different support phrasing; in directive mode tool-backed turns speak the backend directive exactly.

No streaming (LLM mode)

- Baseline raw pipeline. Tool-backed turns can require one model call to choose the tool and a second model call to verbalize the tool result.
- This run showed one high-latency callback turn, visible in the LLM p95.

Pipeline stage	What it measures	Median	P95
End-of-speech detection	VAD/turn detection delay after the user stops speaking.	0.68s	0.69s
Speech-to-text	Time to produce the transcript from captured audio.	1.25s	2.85s
LLM first token	Time until model output begins.	1.09s	1.65s
LLM total response	Total model time for the turn. Tool turns may include planning and post-tool response.	2.53s	8.51s
Local tool chain	Local tool execution and mock-data lookup time.	0.00s	0.01s
TTS first audio	Time until backend receives first audio bytes from TTS.	1.23s	1.71s
TTS complete synthesis	Time until the full response audio is generated.	2.14s	2.93s
Browser delivery	Browser-side response handling and playback-start overhead.	0.04s	0.45s
Total end-to-heard	Total time from user finishing speech to the browser starting response playback.	6.61s	13.95s

No streaming (LLM mode) - interaction transcript

Agent replies are reconstructed from the observed conversation flow because the CSV records user transcripts and latency fields but not assistant text.

#	User said	Agent replied	Tool?	End-to-heard
1	Hello	Hi there! How can I assist you today?	No	6.61s
2	What can you do for me?	I can help with order status, ticket status, and arranging a support callback.	No	4.96s
3	I need help with my order status.	May I have your full name so I can look up your order?	Yes	6.92s
4	My name is Arjun Mehta.	Thanks, Arjun. May I have your order ID?	Yes	8.16s
5	The order ID	I did not catch the full order ID. Please say the digits one at a time.	Yes	5.63s
6	The order ID is 2205.	I could not find that order for your account. I can connect you with a representative or schedule a callback.	Yes	7.48s
7	Can you schedule a callback for me?	Sure, what day and time works best for the callback?	Yes	6.11s
8	Please schedule it for tomorrow after 7 PM.	Your callback is scheduled for October 1, 2026 at 7:30 PM.	Yes	13.95s
9	Okay, thank you.	You're welcome. Have a great day.	No	5.80s

No streaming (directive mode)

- Optimization run. Tool-backed turns still use the LLM to choose tools, but final speech comes directly from the backend directive.
- This removed the post-tool model generation pass and made tool outcomes deterministic.

Pipeline stage	What it measures	Median	P95
End-of-speech detection	VAD/turn detection delay after the user stops speaking.	0.68s	0.69s
Speech-to-text	Time to produce the transcript from captured audio.	1.37s	4.35s
LLM first token	Time until model output begins.	1.46s	2.32s
LLM total response	Total model time for the turn. Tool turns may include planning and post-tool response.	1.77s	2.67s
Local tool chain	Local tool execution and mock-data lookup time.	0.00s	0.32s
TTS first audio	Time until backend receives first audio bytes from TTS.	1.30s	2.54s
TTS complete synthesis	Time until the full response audio is generated.	1.85s	3.07s
Browser delivery	Browser-side response handling and playback-start overhead.	0.04s	0.12s
Total end-to-heard	Total time from user finishing speech to the browser starting response playback.	6.19s	8.76s

No streaming (directive mode) - interaction transcript

Tool-backed agent replies use deterministic backend directive strings; non-tool greetings are reconstructed from the call flow.

#	User said	Agent replied	Tool?	End-to-heard
1	Hello.	Hi there! How can I assist you today?	No	8.76s
2	What can you do for me?	I can help with order status, ticket status, and arranging a support callback.	No	6.15s
3	I needed assistance with an update on my order.	I can help with that. May I have your full name?	No	7.64s
4	Hello.	Hi there! How can I assist you today?	No	6.47s
5	My name is Arjun Mehta.	May I have your order ID?	Yes	5.72s
6	The order ID is 2205.	I couldn't find order 2, 2, 0, 5 for this account. Would you like me to set up a callback?	Yes	6.19s
7	Yes please.	What day and time would work best for a callback?	Yes	6.78s
8	Tomorrow after 6 pm.	Your callback is scheduled for October 1, 2026 at 6:30 PM.	Yes	4.78s
9	Okay, thank you.	You're welcome. Have a great day.	No	4.29s